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Laramide Orogenic Fluid Flow into the Western Canada Sedimentary Basin: Evidence from Paleomagnetic Dating of the Kicking Horse Mississippi Valley-Type Ore Deposit

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Abstract

The Kicking Horse and Monarch Mississippi Valley-type zinc-lead ore deposits are in the southern Rocky Mountains of the foreland belt in the Canadian Cordillera of British Columbia. These epigenetic sphalerite-galena deposits are hosted by a massive sparry dolomite sheet in Middle Cambrian dolostones of the Cathedral Formation on the southwestern margin of the western Canada sedimentary basin. These strata were folded, thrust faulted, and uplifted during the Laramide orogeny. Paleomagnetic analysis was done on 316 specimens from 32 mineralized and host-rock sites, using alternating field and thermal step demagnetization and isothermal remanence methods. The results show that the characteristic remanent magnetization is carried by both pyrrhotite and magnetite. The increase in characteristic remanent magnetization intensity near Mississippi Valley-type mineralization ties the magnetization to the mineralization event. Fold tests on a small scale (~10 m) and a large scale (~10 km) show that the characteristic remanent magnetization postdates the main Laramide deformation event. Comparison of the characteristic remanent magnetization directions shows that the host rocks were remagnetized when the mineralization was emplaced with a primary characteristic remanent magnetization. The Kicking Horse mineralization and adjacent host rocks form the bulk of the collection and give a Late Cretaceous pole position of 64.0° N, 152.8° W (semi-axes of oval of 95% confidence: δ_p = 5.4°, δ_m = 5.8°; number of sites (specimens) = 19 (180)). The age of this pole coincides with the Cretaceous normal superchron and conforms to the finding of all normal characteristic remanent magnetization directions except for one sample with an antipodal direction. No site retains a characteristic remanent magnetization that predates the Laramide orogeny . The evidence for and against a pre-Laramide Mississippi Valley-type mineralizing and dolomitizing event in the western Canada sedimentary basin is reviewed, and it is concluded that the main fluid flow event in the basin occurred during the Late Cretaceous-Paleocene Laramide orogeny.

Introduction

OVER the past decade the timing of the fluid flow event(s) causing dolomitization in the western Canada sedimentary basin has been debated with some researchers favoring a Cretaceous-Paleocene flow related to the Laramide orogeny (e.g., Garven, 1985; Qing and Mountjoy, 1992) and others arguing for an earlier age, usually mid to late Paleozoic, from a pre-Laramide flow (e.g., Nesbitt and Muehlenbachs, 1991; Yao and Demicco, 1995). Oliver (1986, 1992) noted the relationship on a continental scale between orogenesis and the spots and stains of the related fluid flows that include regional dolomitization, transport of petroleum into reservoirs, and the formation of Mississippi Valley-type zinc-lead-barite deposits. These features are difficult to date radiometrically because they commonly lack suitable minerals in sufficient concentrations. However, paleomagnetic methods have been used to date nine major Mississippi Valley-type deposits in North America and shown in every case that they are coeval with orogenesis in the surrounding or adjacent orogen (Symons et al., 1996). This paper reports on paleomagnetic results from the Kicking Horse and Monarch Mississippi Valley-type deposits. They are the only Mississippi Valley-type deposits in the Canadian Cordillera to have been mined. Direct dating of their dolomite host rocks and mineralization provides a test of the pre-Laramide vs. Laramide argument for orogenic fluid flow in the cordillera and the western Canada sedimentary basin, which is critically examined in the concluding discussion.

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Regional Geology

The study area is on the western side of the southern Rocky Mountains in the foreland belt of the cordillera (Fig. 1a). Here the Phanerozoic supracrustal sequence is a 3- to 5-km-thick wedge that thins northeasterly to its margin across the western Canada sedimentary basin (Price, 1994). In the foreland belt, Laramide compression on east-verging thrust faults has resulted in as much as 200 km of shortening, folding and stacking the sequence in west-dipping thrust sheets with an aggregate thickness of 10 to 15 km (McMechan and Thompson, 1989). Clastic sediments of the latest Upper Proterozoic Miette Group of the Windermere Supergroup form the basement in the study area. Lying unconformably over them are >900 m of quartz arenites and shales of the Lower Cambrian Gog Group (Price et al., 1980).

Following the northwesterly trend of the foreland belt is a Middle Cambrian to Middle Ordovician facies transition which forms a prominent lineament known both as the Kicking Horse rim or Cathedral escarpment (Aitken, 1971; Fritz,

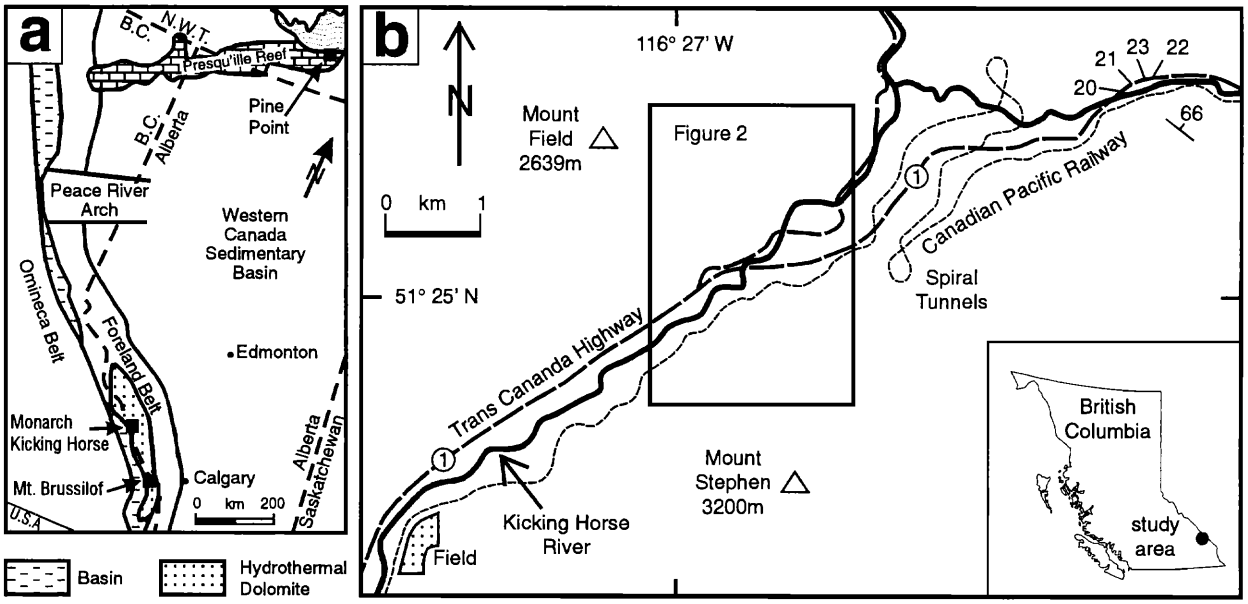


FIG. 1. Location of the study area and of sites 20–23.

1990). It separates deep-water miogeoclinal siliciclastic shales and carbonates of the Chancellor Group to the west from a shallow-water platform carbonate and siliciclastic sequence to the east, and it cuts across northwesterly on the immediate west side of the Mississippi Valley-type mineralization in the Kicking Horse and Monarch mines (Fig. 2a).

At the mines the platform sequence (Fig. 2a) consists from bottom to top of: (1) ~30 m of gray siliciclastic siltstones of the Mount Whyte Formation, (2) ~300 m of massive cliff-forming limestones and dolostones of the Cathedral Formation, (3) ~170 m of greenish fossiliferrous shales of the Stephen Formation, (4) ~800 m of massive dolostones of the

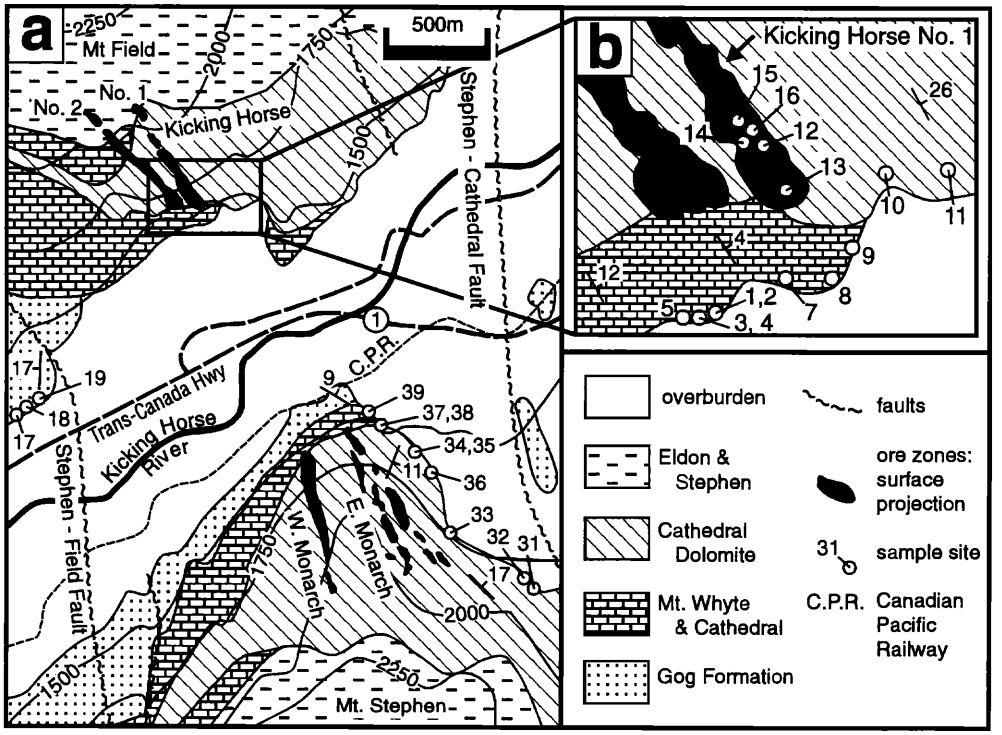


FIG. 2. Geology of the study area (from Ney, 1957, and Price et al., 1980) with the locations of sites 1–19 and 31–39.

Eldon Formation, (5) ~160 m of dark flaggy dolostones over a thin basal siliciclastic shale of the Pika Formation, and (6) ~50 m of green, purple, and red shales of the Arctomys Formation (Price et al., 1980). The platform sequence extends ~4,000 m farther upward in the region, terminating with Upper Triassic strata.

There is little evidence of significant tectonism in the foreland belt from Middle Cambrian to the start of the Laramide orogeny. Laramide uplift in the belt is recorded in three sedimentary assemblages in the adjacent basins (Price, 1981). Initial compressional collision of the allochthonous Intermontane belt terranes and indentation of the western edge of the North American craton occurred from mid-Late Jurassic (~155 Ma) to Early Cretaceous (~130 Ma), forming the Omineca crystalline belt and depositing the Kootenay assemblage (Price, 1994). Between mid-Early Cretaceous (~120 Ma) and early Late Cretaceous (~90 Ma), the Insular belt terranes impacted outboard of the Intermontane terranes to form the Coast Plutonic Complex welt. Right-lateral transpression of these allochthonous terranes against the craton led to rapid uplift in the foreland belt with deposition of the Blairmore assemblage. From mid-Late Cretaceous (~80 Ma) through the Paleocene (~57 Ma), the Brazeau assemblage was deposited. It reflects dominantly dextral strike-slip motion with minor transpression of the allochthonous terranes against the craton. Orogenic uplift in the foreland belt ended by mid-Eocene (~45 Ma) with the onset of right-lateral transtension in the accreted terranes to the west. These changes in Laramide tectonism are thought to coincide with changes in the plate tectonic regime on the west coast of North America (Engebretson et al., 1985; Price 1994).

Extensive hydrothermal dolomite sheets and veins are found along the east side of the Cathedral escarpment for ~250 km (Fig. 1a; Nesbitt and Muehlenbachs, 1994a). Next to the escarpment these sheets and veins have substantially replaced the Cambro-Ordovician strata. Also, the sheets thin progressively and die out over a few tens of kilometers northeastward. This geometry plus northeasterly decreasing fluid inclusion homogenization temperatures from ~200° to ~100°C in the dolomites suggest a northeasterly fluid flow (Yao and Demicco, 1995).

Mine Geology

The Kicking Horse and Monarch mines are in Yoho National Park about 5 km east of Field, British Columbia, on either side of the Kicking Horse Pass valley (Fig. 1b). Their geology has been described by Ney (1951, 1957). The mineralization is found about ~130 m above the base of the Cathedral Formation, close to the contact between an ~20-m-thick black dolostone horizon and an overlying >200-m-thick layer of massive light gray dolostone to coarse white hydrothermal dolomite (Fig. 2a). The several ore zones occur in small amplitude anticlinal flexures near the crest of a broad anticline with dips of 2° to 12° ENE (Figs. 1a and 2b; Cook, 1975). The ore zones are concordant linear bodies that range up to 400 m long, 40 m wide, and 6 m thick. They plunge gently at ~7° ± 7° NNW and may have been continuous across the valley prior to erosion. The mineralization consists of coarsely crystalline white to dark brown sphalerite and galena. It is disseminated in a sparry white dolomite which cements dolostone breccia clasts. Production occurred from 1912 to 1952. Together, both mines yielded about 826,180 metric tons (t), grading ~8.6 percent zinc, ~5.6 percent lead, and ~30.4 g/t silver. Pyrite ranges from trace amounts in mined ore to dominance in some mineralized zones. Traces of chalcopyrite have been found in ore, and traces of quartz have been seen in vugs. From ~500 to ~1,500 m to the east and west of the mines are the major Stephen-Cathedral and Stephen-Field faults, respectively (Fig. 2a). Both are northerly trending normal faults that splay off the Simpson Pass thrust fault and dip toward the mineralized zones with displacements of ~1,000 and ~300 m, respectively, so that the broad anticlinal structure of the mines is in a graben. Several smaller faults parallel these regional faults, and drag folds are common next to them (Cook, 1975; Price et al., 1980).

Experimental Methods

Sampling

A total of 405 oriented core specimens were prepared from 106 block samples from 32 sites (Figs. 1 and 2). Five sites (12–16) were collected from massive to disseminated sphalerite and galena in hydrothermal dolomite from inside the Kicking Horse mine, and an additional six (6–11) from barren dolomite in adits and drill pockets at surface. Sites 1 to 5 were collected for a paleomagnetic fold test (Graham, 1949) from around three limbs of a kink fold with an amplitude of ~6 m in dolomitic limestones of the Cathedral Formation. Another seven sites (31–34, 36, 38, 39) in dolostones and two sites (35, 37) in hydrothermal dolomite were located near the Monarch mine. Finally, seven sites were collected for a regional paleomagnetic fold, three (17–19) in westerly dipping quartzites of the Gog Formation, and four (20–23) were in easterly dipping carbonates of the Cathedral Formation (Figs. 1 and 2).

Measurements

All specimens were stored in a magnetically shielded room for three months to allow their viscous remanent magnetization to decay. Their natural remanent magnetizations were measured using an automated CTF Systems Inc. cryogenic magnetometer with a resolution limit of $\sim 3 \times 10^{-6}$ A/m. From the natural remanent magnetization data, two to four specimens per block with the most intense natural remanent magnetization intensities were chosen for analysis to give 10 specimens for most sites (Table 1), or 312 specimens in total.

First, two specimens per site were alternating field demagnetized in 12 steps to 130 mT, using a Sapphire Instruments SI-4 demagnetizer. Then one specimen per site was thermally demagnetized in eight steps to 560°C using a Magnetic Measurements TD-1 demagnetizer. Based on these results, the remaining specimens were alternating field demagnetized at four intensities between 20 and 70 to 110 mT and then thermally demagnetized at 325°, 375°, or 450°C. Their characteristic remanent magnetization directions were obtained using orthogonal demagnetization diagrams (Zijderveld, 1967) and principal component analysis (Kirschvink, 1980), and the site and unit mean directions were calculated following Fisher

TABLE 1. Site Mean Characteristic Remanent Magnetization

Site	Lithology	Strike, dip (°)	Specimens Nm	Specimens Nu	Demagnetization AF (mT)	Demagnetization Th (°C)	MAD angle Med (°)	MAD angle Max (°)	ChRM direction Dec. (°)	ChRM direction Inc. (°)	ChRM direction α_{95} (°)	k
1	dLst	339, 16	10	9	20–110	130–530	1.5	4.8	306.2	76.8	5.9	77
2	dLst	328, 16	10	9	20–110	70–500	1.6	3.5	12.4	79.1	4.2	153
3	dLst	149, 65	10	7	20–130	70–450	1.6	2.5	326.8	77.7	5.1	143
4	dLst	329, 26	10	9	20–130	70–500	1.9	2.7	302.7	80.4	3.2	259
5	dLst	327, 23	10	10	20–130	130–500	2.3	4.8	307.4	72.6	5.6	76
6	hDol	325, 04	10	10	20–130	130–325	4.4	9.4	317.8	82.5	6.4	58
7	hDol	325, 06	10	10	20–130	250–325	1.8	7.3	46.3	66.7	6.3	59
8	hDol	325, 08	10	9	20–130	130–500	2.2	3.0	315.0	81.4	5.6	86
9	bDst	330, 10	10	10	20–130	250–450	2.0	3.4	314.3	79.4	6.4	58
10	hDol	330, 15	10	10	20–130	130–325	2.0	6.7	341.0	70.8	10.1	24
11	hDol	330, 26	10	9*	20–130	130–250	1.7	8.1	289.8	79.1	8.1	42
12	Min	325, 08	10	10	20–130	130–450	2.1	3.4	283.5	76.2	3.8	160
13	Min	325, 08	10	10	30–130	70–450	2.4	5.6	282.7	74.9	2.4	409
14	Min	325, 08	9	9	20–130	250–325	2.2	2.5	357.9	85.6	5.5	90
15	Min	325, 08	10	10	20–130	70–325	3.8	9.2	348.1	68.7	4.8	105
16	Min	325, 08	10	10	20–130	70–325	1.7	3.2	303.1	77.5	4.2	137
17	Qzt	135, 30	9	9	20–70	70–500	7.8	13.6	97.5	71.8	64.5	1.6
18	Qzt	155, 24	10	9	20–130	70–450	7.7	9.4	332.6	71.1	16.2	11.0
19	Qzt	175, 18	10	10	20–70	130–450	6.0	9.1	325.7	83.9	42.6	2.2
20	Dst	305, 75	10	9	20–130		1.5	5.5	358.2	69.1	7.5	49
21	dLst	305, 65	10	10	20–130	130–325	4.9	10.8	314.3	76.8	6.9	51
22	Dst	320, 62	10	10	20–130	70–325	2.0	4.2	303.8	76.8	3.2	224
23	bLst	304, 64	10	10	20–130	70–325	1.6	4.1	286.5	75.1	3.8	162
31	hDol	315, 18	10	10	20–130	70–325	1.6	2.7	324.8	84.3	6.3	60
32	hDol	315, 16	8	8	30–130	70–325	1.9	3.3	303.1	72.5	9.9	33
33	hDol	335, 10	10	9	20–130	130–250	1.4	2.9	85.9	80.7	8.3	39
34	Dst	172, 04	10	10	20–130	130–375	1.6	2.6	344.7	86.1	4.7	105
35	hDol	190, 06	6	6	20–130	70–375	4.0	5.5	261.2	26.1	53.5	2.5
36	Dst	215, 09	10	10	20–130	70–450	1.4	2.5	343.2	86.7	3.7	173
37	hDol	300, 08	10	10	20–130	250–450	1.2	2.8	123.8	84.0	8.2	36
38	Dst	300, 08	10	10	20–130	130–450	2.6	6.7	347.4	84.5	6.6	54
39	Dst	300, 09	10	10	20–110	70–250	2.1	2.9	220.6	88.9	3.5	196

Notes: lithology: b = black, d = dolomitic, Dol = dolomite, Dst = dolostone, h = hydrothermal, Lst = limestone, Min, = sphalerite mineralization, Qzt = quartzite; strike: by azimuth, dip: right-handed convention; specimens: Nm = number measured, Nu = number used for calculation; demagnetization range represented: alternating field (AF) in millitesla (mT); thermal (Th) in degrees Celsius (°C); maximum angular deviation (MAD) angle of Kirschvink (1980) giving the median (med) and maximum (max) values in degrees of arc (°); characteristic remanent magnetization (ChRM) direction given by declination (Dec.), inclination (Inc.), and radius of cone of 95 percent confidence (α_{95}) and precision parameter (k) of Fisher (1953); asterisk denotes the presence of one antipodal reversed direction

(1953). The magnetic mineralogy of 14 representative specimens was examined using saturation isothermal remanent magnetization analysis by magnetizing them in 11 direct field steps up to 900 mT and then alternating field demagnetizing the saturation isothermal remanent magnetization in six steps to 130 mT.

Results

Natural remanent magnetization

The 210 specimens from the Kicking Horse mine area are ~50 m on average from known mineralization and have a median natural remanent magnetization intensity of 1.1×10^{-3} ($Q_1 = 5.9 \times 10^{-4}$, $Q_3 = 1.9 \times 10^{-3}$) A/m (median (lower quartile, upper quartile)) with similar median values of 1.3, 1.1, and 1.1×10^{-3} A/m for the mineralized, hydrothermal dolomite, and dolomitic limestone specimens, respectively. These values are more intense than those found in most Mississippi Valley-type deposits to date (Symons and Sangster, 1994). The 106 specimens from the Monarch mine area are ~500 m on average from known mineralization and give a lower median value of 8.6×10^{-4}($Q_1 = 4.8 \times 10^{-4}$, $Q_3 = 1.6 \times 10^{-3}$) A/m. The weakest specimens, collected for the regional fold test, are ~6.5 km on average from known mineralization and give median intensities 3.3, 2.4, and 2.3×10^{-4} A/m for the 33 dolostone, 26 dolomitic limestone, and 30 quartzite specimens, respectively. The drop in natural remanent magnetization intensity away from the center of Mississippi Valley-type mineralization is statistically significant and suggests that the magnetization is related to the mineralization.

Thermal demagnetization

Fourteen of the 29 carbonate specimens which were thermally step demagnetized in detail show an average drop of 6 percent in natural remanent magnetization intensity by the 70°C step, indicating the presence of relatively little viscous remanent magnetization. In contrast, the remaining 15 specimens show significant viscous remanent magnetization with initial intensity drops down to as low as 13 percent of the

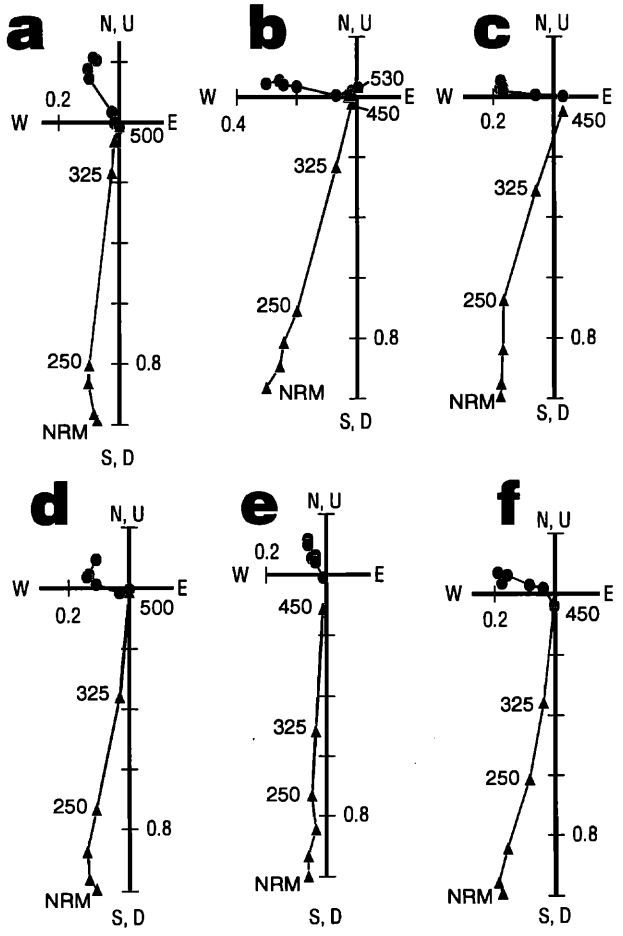


FIG. 3. Orthogonal thermal step demagnetization plots of example specimens from sites (a) 2, (b) 23, (c) 12, (d) 34, (e) 36, and (f) 13. Projection in the horizontal north-east-south-west (N, E, S, W) and vertical up-east-down-west (U, E, D, W) planes are shown as circles and triangles, respectively. The natural remanent magnetization (NRM) direction and the temperature (°C) of some step directions are shown. The axial intensities are expressed as a ratio of the natural remanent magnetization intensity.

natural remanent magnetization intensity with a median drop to 49 percent. At higher temperatures, nearly all of the carbonate specimens define characteristic remanent magnetization directions which are directed steeply downward to the northwest. None show marked intensity drops in the 70° to 130°C range, indicating that goethite is not a carrier of the characteristic remanent magnetization. Most specimens, however, do show a knee or marked drop in the 250° to 325°C temperature range which indicates the presence of pyrrhotite (Fig. 3 a and b). Several of these specimens show the characteristic remanent magnetization direction up to about 530°C, indicating that magnetite is also present (Fig. 3 c and d). A few specimens display a knee above 325°C but below 585°C, suggesting that only magnetite is present (Fig. 3a). Finally a few carbonate specimens show a simple linear intensity decrease to about 450°C which is inconclusive in terms of the characteristic remanent magnetization carrier (Fig. 3f). Nevertheless, this is only the second Mississippi Valley-type deposit of the 12 reported on to date in which pyrrhotite is shown to be a significant characteristic remanent magnetization carrier (Symons et al., 1997a). The three quartzite specimens show linear drops to about 500°C, indicating the presence of magnetite with an increase thereafter, indicating that new magnetic minerals were being formed by oxidation in the oven.

Alternating field demagnetization and site mean directions

Regardless of lithology, nearly all carbonate specimens behaved similarly on alternating field step demagnetization. Up to ~20 mT, a viscous remanent magnetization is removed which is directed northward and fairly steeply downward, and it is thought to be recording the present Earth's magnetic field direction (Fig. 4). From ~20 mT and up to the 110 to 130 mT step, a characteristic remanent magnetization is isolated in most cases, which is directed steeply downward to the northwest and decays to the origin over the entire coercivity spectrum. Table 1 shows well-defined characteristic remanent magnetization directions with a median maximum angular deviation angle (Kirschvink, 1980) of 1.9° (Q_1 = 1.6°, Q_3 = 2.2°), with a maximum of only 10.8° for all 273 specimens.

After an alternating field demagnetization of 20 mT, the decrease in remanence intensity of the carbonate specimens remains obvious when going away from the known Mississippi Valley-type mineralization. The median intensities are 5.6, 4.5, and 1.7 × 10^{-4} A/m at average distances of ~50, ~500, and ~6,500 m from known Mississippi Valley-type mineralization for 145, 75, and 38 carbonate specimens, respectively. The most basic observation is that dolomitization is most intense in the Mississippi Valley-type zones and becomes progressively less intense away from the mineralization, with a lower magnetite and pyrrhotite content as a consequence. Indeed, dolomitization over the Kicking Horse and Monarch mines provides such a spectacular exposure that locals called it "the Beehive" (Ney, 1951).

The majority of carbonate specimens was subjected to one step of thermal demagnetization, usually 325°C, after alternating field step demagnetization to 110 mT (Fig. 4g). For the majority, the characteristic remanent magnetization was well defined by the alternating field steps and the residual remanence was erased by the thermal step, again indicating that pyrrhotite is an important remanence carrier. Except for site 35, which is excluded from further discussion, the carbonate sites have well-clustered specimen characteristic remanent magnetization directions with a median for their cone of 95 percent confidence (α_{95}; Fisher, 1953) of 5.5° (Q_1 = 3.8°, Q_3 = 6.9°). In contrast the quartzite specimens retain a low stability remanence in the 20- to 50-mT range, which gives poorly defined characteristic remanent magnetization directions. These directions fail to form coherent clusters, giving site mean characteristic remanent magnetization directions with cones of 95 percent confidence of >15° and low to random precision parameter (k) values (Table 1). For this reason the directions from the Gog quartzites are not considered further.

Saturation isothermal remanent magnetization

The saturation isothermal remanent magnetization analyses support the thermal step demagnetization data in showing

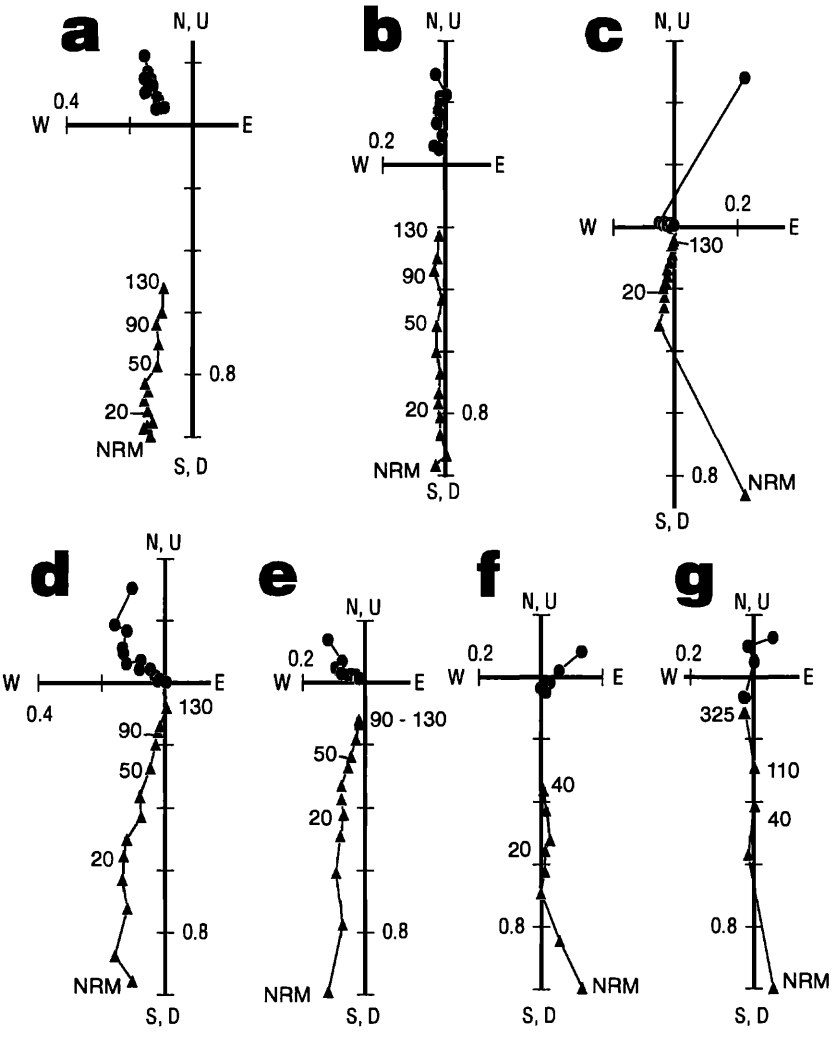


FIG. 4. Orthogonal alternating field step demagnetization plots for example specimens from sites (a) 3, (b) 9, (c) 12, (d) 15, (e) 22, (f) 37, and (g) 11. Conventions as in Figure 3 except that some step values are labeled in millitesla (mT).

that pyrrhotite is the dominant characteristic remanent magnetization carrier in the carbonate specimens. Alternating field step demagnetization of the saturation isothermal remanent magnetization of the Kicking Horse mine specimens gives decay curves which indicate single domain or pseudosingle domain magnetite or pyrrhotite (Fig. 5b). However, their saturation isothermal remanent magnetization acquisition curves (Fig. 5a) climb too slowly to be dominantly magnetite but do climb at rates that are typical for natural single domain or pseudosingle domain pyrrhotites except for the site 7 specimen. Similarly, the saturation isothermal remanent magnetization curves for the carbonate specimens from the Monarch mine area and regional fold test group are indicative of single domain and pseudosingle domain pyrrhotite except for the site 20 specimen, which is an unusual pink dolostone which gives curves that are typical for fine-grained hematite (Fig. 5c and d). Finally, the Gog quartzite specimen from site 18 also gives curves which indicate that pseudosingle domain pyrrhotite is the characteristic remanent magnetization carrier.

Statistical Analysis

The site mean characteristic remanent magnetization directions from the Kicking Horse mine form a coherent cluster except for the aberrant mean from site 7 (Table 1, Fig. 6a). Its explanation is uncertain; however, possibilities include a lightning strike, an unrecognized fault, etc. Nevertheless it is very much an outlier relative to the Kicking Horse population, gives an impossible direction geologically, and, therefore, is excluded from the hydrothermal dolomite group. Next, the five site means from dolomitic limestone on the limbs of the 6-m amplitude kink fold (sites 1–5) were averaged with and without tilt correction (Table 2, entries 3 and 4). Tilt correction scatters the mean directions increasing the α_{95} value from 6.2° to 36.2° and decreasing the precision parameter (k) value from 155.8 to 5.4. Using the paleomagnetic fold test of McFadden and Jones (1981) with the block mean directions for the five sites gives a test statistic of 5.3331, which is very much greater than the comparison statistic of 0.5038 for probability p = 0.005. This test shows with $\geqslant$99.5 percent

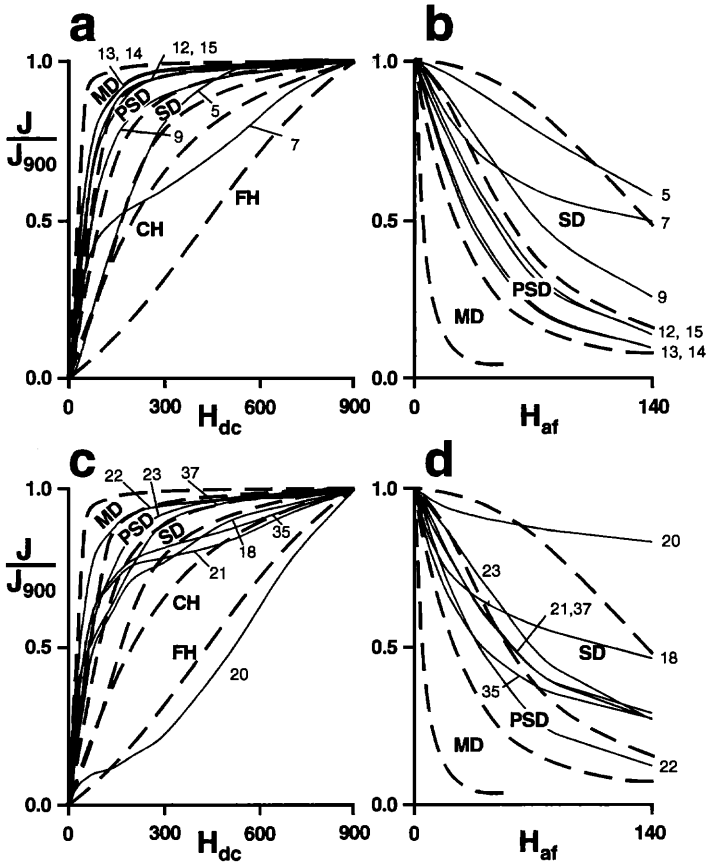


FIG. 5. Saturation isothermal remanent magnetization curves for test specimens which are identified by their site number. J/J_{900} is the ratio of the measured intensity to saturation isothermal remanent magnetization intensity. H_{dc} and H_{af} are the intensities in mT of the magnetizing direct and demagnetizing alternating fields. Type curves are shown for multidomain (MD), pseudosingle domain (PSD), and single domain (SD) magnetite, and for coarse-grained (CH) and fine-grained (FH) hematite.

confidence that the remanence is postfolding in origin. Therefore the site mean directions, uncorrected for tilt, were calculated for the mineralization (sites 12–16) and hydrothermal dolomite (sites 6, 8–11; Table 2, entries 1 and 2). The salient feature is that the mean directions for the dolomitic limestones, dolomites, and mineralization (Fig. 6b) are all within ~1.5° of each other, compared to α_{95} values between 6.2° and 9.2°, meaning that the three populations can certainly be drawn from a common single population and, therefore, averaged to form a single mean direction of D = 317.0°, I = 78.7° (α_{95} = 3.3°, k = 131.7°; declination, inclination (cone of 95% confidence, precision parameter)).

The mean directions from sites in the steeply dipping dolomitic limestones and dolostones at the highway cut (sites 20–23) can be used for a regional fold test by comparing them to the dolomitic limestone sites at the Kicking Horse mine (Table 2, entries 6–9). Again, the fold test of McFadden and Jones (1981) shows overwhelmingly with ⩾99.5 percent confidence that the remanence is postfolding in origin with a test statistic of 26.694 ⩾ 0.9392 for p = 0.005 for the two populations that have comparable precisions before unfolding. When the highway cut population is compared to the overall Kicking Horse mine population without tilt correction (Table 2, entries 5 and 6) using the test of McFadden and Lowes (1981) for populations with comparable precisions, the calculated statistic (F_c) of 0.0207 is very much less than the theoretical statistic (F_t) of 0.1927 for p = 0.05. This indicates with 95 percent confidence that the two populations can be drawn from a single population and, therefore, they have been combined to give D = 317.6°, I = 78.2° (α_{95} = 3.1°, k = 121.6; Table 2, entry 10).

The dolostone and hydrothermal dolomite sites from the Monarch mine (sites 31–34, 36–39) yield, in general, steeper characteristic remanent magnetization mean directions than those found for the Kicking Horse mine. Using the McFadden and Lowes (1981) test, F_c = 0.3408 > 0.2870 = F_t for p = 0.005, showing with >99.5 percent confidence that the Kicking Horse and Monarch populations cannot be drawn from a single population. This means that they cannot be validly combined into one population so that they require differing explanations.

Pole Positions

The pole position obtained for the Kicking Horse mine mineralization, sparry dolomite, and remagnetized limestone host rocks (Table 2, entry 5) is located at 63.5° N, 151.0° W

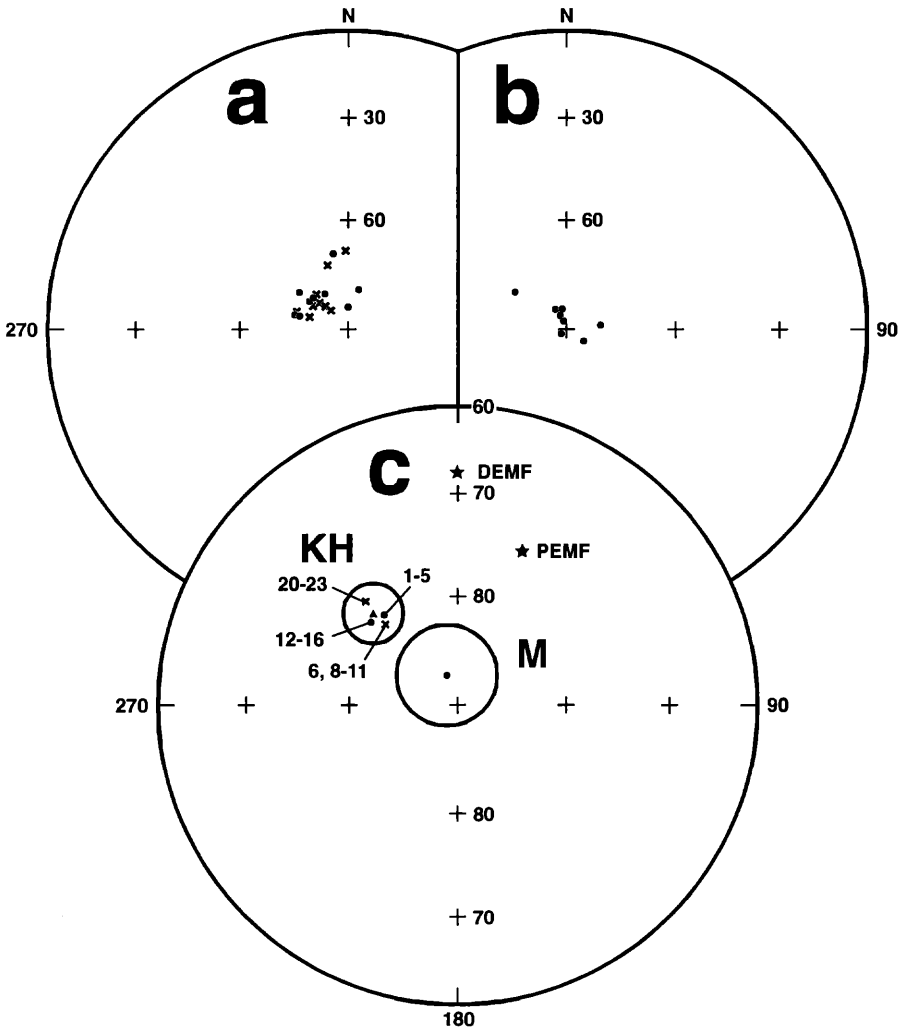


FIG. 6. Lower hemisphere equal area stereograms showing mean characteristic remanent magnetization directions. a. Host rock (●) and mineralized zone (x) sites from the Kicking Horse mine area. b. Sites from the Monarch mine area. c. Shown an enlarged polar segment of a stereogram with the overall Kicking Horse (KH,▲) and Monarch (M, ●) characteristic remanent magnetization mean directions surrounded by their cones of 95 percent confidence. Also shown are the mean directions for groups of sites (Table 2) and the present and average Quaternary poles for the earth's magnetic field direction (PEMF and DEMF, respectively).

TABLE 2. Group Mean Remanence Directions

Group	Sites	Cor.	N	Declination (°)	Inclination (°)	α_{95} (°)	k
1. Kicking Horse mineralization	12–16	U	5	311.7	78.4	9.2	70.6
2. Kicking Horse dolomite	6, 8–11	U	5	319.2	79.2	5.7	179.3
3. Kicking Horse limestone	1–5	U	5	302.3	78.4	6.2	155.8
4. Kicking Horse limestone	1–5	C	5	325.5	77.2	36.2	5.4
5. Kicking Horse all	1–6, 8–16	U	15	317.0	78.7	3.3	131.7
6. Highway cut carbonates	20–23	U	4	319.5	76.3	10.5	77.3
7. Highway cut carbonates	20–23	C	4	24.0	20.4	19.4	23.5
8. Regional fold test	1–5, 20–23	U	9	319.9	77.5	4.7	121.1
9. Regional fold test	1–5, 20–23	C	9	12.9	52.6	30.2	3.9
10. Kicking Horse + highway cut	1–6, 8–16, 20–23	U	19	317.6	78.2	3.1	121.6
11. Monarch dolostones	31–34, 36–39	U	8	342.0	87.0	5.5	103.6

Notes: Correction of characteristic remanent magnetization directions (Cor.): uncorrected (U) or tilt corrected (C); N = the number of site mean directions averaged; other abbreviations as in Table 1

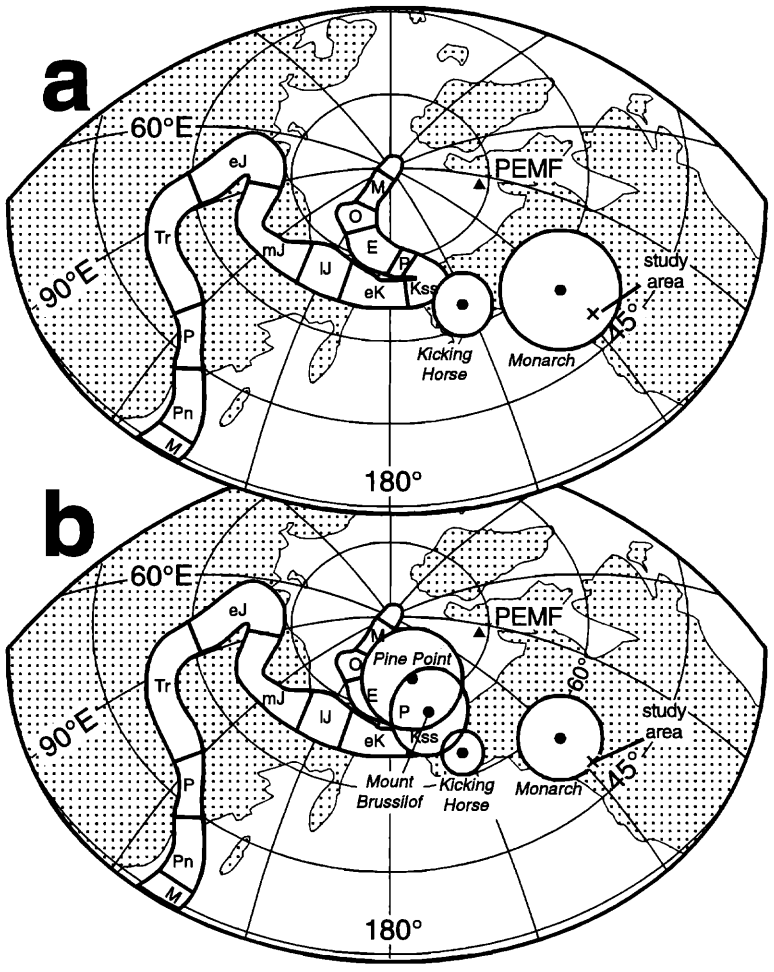


FIG. 7. Pole positions for (a) the Kicking Horse (KH) and Monarch (M) mine areas with their ovals of 95 percent confidence, and (b) the Kicking Horse and other mine areas in the western Canada sedimentary basin with their circular standard deviation ovals. The apparent polar wander path from the Missippian to present is from Van der Voo (1990, 1993).

(δ_p = 6.0°, δ_m = 6.3°; latitude, longitude (semiaxis of oval of 95% confidence)). Adding the data from the four highway cut sites moves the pole insignificantly to 64.0° N, 152.8° W (δ_p = 5.4°, δ_m = 5.8°). This best estimate pole is in present-day Alaska and overlaps onto the Cretaceous stillstand portion from ~133 to ~66 Ma of the North American apparent polar wander path of Besse and Courtillot (1991; Fig. 7). The detailed and regional fold tests show that this remanence postdates Laramide folding which agrees with the structural evidence from mine mapping, and the pole position confirms a Late Cretaceous to Paleocene age for the genesis of the Kicking Horse Mississippi Valley-type mineralization. The pole for the Monarch mine area is at 57.0°N, 119.8°W (δ_p = 10.9°, δ_m = 10.9°). Thus it falls well off the apparent polar wander path for North America for any time since the Middle Cambrian host rocks were deposited, but it is closest to the Cretaceous stillstand. We attribute its offset from the apparent polar wander path either to minor (~9°) postdolomitization tilt of the sampled rock face, where drag folding would shallow the dip appropriately on the adjacent Stephen-Cathedral normal fault, or simply to an insufficient number of sites being averaged. Evidence for minor postdolomitization and postmineralization motion has been noted from mine mapping (Ney 1951, 1957). It can also be seen in the differing trends of the mineralized zones in the same Laramide anticline on either side of the Kicking Horse River (Fig. 2).

Statistical Test of Pole Position

Recently, Lewchuk and Symons (1995) noted that Bingham (1974) statistics provided a better fit to the streaked population of site mean directions commonly found for Mississippi Valley-type mineralization and dolomites than do Fisher (1953) statistics. Bingham statistics fit an error ellipsoid to an elongated distribution, thus achieving a good fit with relatively small error limits, whereas Fisher statistics try to force-fit an error circle that results in a poor fit with a relatively large error limit. Lewchuk and Symons (1995) argued that the elongation reflected apparent polar wander during emplacement of the Mississippi Valley-type mineralization which they estimated to have taken about 10 m.y. on

average. If they are correct, then the Kicking Horse ore deposit represents the opposite condition because its remanence formed during the Cretaceous stillstand when there was no apparent polar wander and, therefore, it should yield a circular distribution within statistical error. In fact, this is the case both in direction space and pole space because the bipolar parameters of 1.59 and 1.78 are very much less than the girdle parameters of 402 and 121, respectively (Bingham 1974). Indirectly, this test confirms that apparent polar wander is normally recorded in the genesis of Mississippi Valley-type deposits, as proposed by Lewchuk and Symons (1995).

Origin of the Remanence

The characteristic remanent magnetization of the Mississippi Valley-type ore and its host rocks is chemical in origin. In other Mississippi Valley-type deposits the characteristic remanent magnetization is shown to be carried by magnetite and pyrrhotite in the dolomite as well as in sphalerite and galena crystals (Symons et al., 1996, 1997a). Fluid inclusion data (Nesbitt and Muehlenbachs, 1995a), apatite fission track thermochromometry (Ravenhurst et al., 1994), and metamorphic petrology and conodont alteration index values (Greenwood et al., 1991) reflect burial metamorphism of only the lowest prehnite-pumpellyite facies, which is an insufficient temperature to reset the characteristic remanent magnetization of single domain or pseudosingle domain pyrrhotite or magnetite (Pullaiah et al., 1975; Dekkers, 1989; Dunlop, 1995); therefore, both minerals were nucleated and grown in Laramide time. This could happen in an open system in which fluid flow caused massive replacement and open-space filling with substantial chemical changes such as the conversion of primary limestone to hydrothermal dolomite with the concurrent precipitation of new pyrrhotite and magnetite. The statistically significant increase in characteristic remanent magnetization intensity in and near the Kicking Horse Mississippi Valley-type mineralization relative to more remote locations suggests that substantial replacement did occur. Alternatively, the pyrrhotite and magnetite could have nucleated in a substantially closed system. If the preexisting host rocks were recrystallized to convert dolostones to dolomite through pressure solution transfer, or Riecke's principle, in which points of high stress in a crystal are dissolved and reprecipitated at adjacent sites of lower stress on the crystal edges at temperatures below ~200°C (Twiss and Moores, 1992). This process would result in pyrrhotite and magnetite inclusions in the crystals being dissolved and reprecipitated and would result in an increase in dolomite crystal size along with decreased porosity and permeability. Geochemical evidence favoring this process is given later.

Discussion

The paleomagnetic results show that both the Mississippi Valley-type mineralization and dolostone host rocks carry a Laramide characteristic remanent magnetization. The following discussion leads to the conclusion that the characteristic remanent magnetization is primary in the mineralization, and that it is either primary due to dolomitization or secondary due to pressure remagnetization in the host rocks. Arguments are also made to counter evidence that has been cited in favor of a pre-Laramide age for both the dolomitization and Mississippi Valley-type mineralization.

Stratigraphy

Nesbitt and Muehlenbachs (1994a, 1995a) noted that Mississippi Valley-type mineralization in the southern Rocky Mountains is found in Middle Cambrian to Late Devonian carbonate strata. Also, the Mississippi Valley-type mineralization at Pine Point on the northeastern side of the western Canada sedimentary basin occurs in Middle Devonian carbonates. Therefore, the mineralization must be younger than late Late Devonian unless there have been multiple mineralizing events and there is no supporting evidence for this option.

Structure

Westervelt (1979), Nesbitt and Muehlenbachs (1991, 1994a), and Yao and Demicco (1995) have argued that the dolomite sheets must predate the Late Jurassic because they were folded and faulted by the Laramide orogeny. Yao and Demicco (1995) identified an early dark brown to black dolomite which replaced primary limestone and a later sparry white dolomite which replaced both primary limestone and the dark dolomite. They considered that both dolomites must postdate the Late Ordovician because Upper Ordovician limestone blocks slid down an Ordovician escarpment to the west of the Cathedral escarpment and have been preserved in a pristine condition in the Chancellor Formation, and that they predate the Late Devonian because boulders of the Middle Cambrian Arctomys Formation were dolomitized prior to deposition on a sub-Devonian unconformity. Their time window is inconsistent with the stratigraphic evidence noted above. Also E.W. Mountjoy (pers. commun.) notes that Arctomys' clasts have a syndepositional sabka texture which is very different from the Middle Cambrian carbonates which were replaced by hydrothermal dolomite, invalidating their latter argument.

Nesbitt and Muehlenbachs (1995a) identify an "early" dolomite that includes the hydrothermal dolomite sheets, the talc and magnesite deposits including Mount Brussilof, and the Mississippi Valley-type mineralization including the Kicking Horse, Monarch, and Pine Point deposits. This population is equivalent to the later dolomites of Yao and Demicco (1995). Nesbitt and Muehlenbachs' (1995a) "late" population includes dolomite and quartz veins which follow preexisting thrust faults or form undeformed near-vertical veins that crosscut folded strata. They consider that the early and late populations record pre-Laramide, probably the late Paleozoic Antler orogeny, and syn- to post-Laramide fluid flow events, respectively.

Based on extensive mapping in the Kicking Horse and Monarch mines, Goranson (1937), Brown (1948), and Ney (1951, 1957) record that the sphalerite-galena ore follows brecciated zones in the cores of minor anticlinal and synclinal folds in the crest of a regional Laramide anticline to form cigar-shaped mineralized zones. They note further that Laramide fractures reflect mostly preore motion with only slight postore movement and that tongues of mineralization and dolomitization extend along these Laramide fractures. This

is powerful evidence for a late or post-Laramide age for both the early white hydrothermal dolomite population and the late dolomite-quartz veins. Another example is the Oldman Mississippi Valley-type showing in carbonates of the upper Upper Devonian Palliser Formation of southwestern Alberta (Holter, 1977). Its Mississippi Valley-type mineralization is in dark and white coarsely crystalline dolomite along a tear fault that splays off the Lewis thrust, about 60 m above it. Further, because the Lewis thrust overlies Upper Cretaceous sedimentary rocks deposited during magnetic polarity chrons 33 N and 34 N at ~77 ± 3 Ma (Lerbekmo and Coulter, 1985; Cande and Kent, 1995; Jerzykiewicz et al., 1996) and apatite fission track data indicate that the main motion on the fault occurred at ~75 Ma (Osadetz et al., 1996a), the Oldman Mississippi Valley-type mineralization postdates significant Laramide deformation.

As described by Nesbitt and Muehlenbachs (1995a), the late syn- to post-Laramide dolomite-quartz veins are either undeformed or little deformed and found in dilated thrust faults or near-vertical fractures. Volumetrically they represent very little dolomite compared to the dolomite sheets. These features suggest that they are terminal to post-Laramide tensional features related to either the Eocene regional extensional regime or to minor post-Laramide isostatic uplift adjustments. If the dolomite sheets are truly pre-Laramide as Nesbitt and Muehlenbachs (1995a) propose, this implies improbably that the Paleozoic platform carbonate sequence was essentially unaffected by the Laramide orogenic fluid flow and compressional deformation events.

The hypothesis of a pre-Laramide genesis for the dolomite sheets is founded on the assumption that they have been folded (Westervelt, 1979). Stated simply, epithermal dolomite can replace folded or unfolded strata with equal ease. The critical factors are a source for magnesium, and the degree of chemical reactivity and the porosity and permeability of a given stratigraphic unit to the fluid flowing through it. The midcontinent of the United States provides a good example of comparable scale to the western Canada sedimentary basin. Although Mississippi Valley-type mineralization is found in minor amounts throughout the Cambrian to Pennsylvanian carbonate platform sequence, the major Pb-Zn-Ba deposits are found in both folded and unfolded dolostones of Late Cambrian to Middle Ordovician age in the eastern Tennessee, central Tennessee, southeastern Missouri, central Missouri, Upper Mississippi Valley, and northern Arkansas districts. Most of these deposits have been dated as late Pennsylvanian or Permian by paleomagnetism and by a variety of radiometric methods, coincident with the Alleghenian and Ouachitan orogenies (Symons et al., 1996). Other clear examples of postdeformational Mississippi Valley-type ore and dolomitization abound, such as at Polaris in the Arctic Archipelago (Symons and Sangster, 1992) and in Silesia (Symons et al., 1995).

Geochemistry

Nesbitt and Muehlenbachs (1991, 1994a, 1995a) have reported results from a large number of fluid inclusion and isotopic analyses for dolomites in the southern Rocky Mountains. Their early and late dolomite populations both give $\delta^{13}C$ values ranging between −3 and 1 per mil, substantially overlapping fluid inclusion homogenization temperature ranges of 100° to 200°C and of 90° to 150°C, respectively, and overlapping $\delta^{18}O$ values ranging between 11 to 18 and 16 to 22 per mil PDB, respectively. These values favor an evolving fluid chemistry rather than fluids from two different source terranes. They also report δD values of −40 to −90 per mil SMOW for their early dolomite and Mississippi Valley-type ores, as low as −140 per mil SMOW for their early talc and magnesite ores, and −120 to −150 per mil SMOW for their late dolomite-quartz veins. Rather than considering this to be a continuum from an evolving fluid, they argue that the early dolomite and Mississippi Valley-type ores are different from the late dolomite-quartz veins, requiring hydrothermal pre-Laramide and meteoric syn- to post-Laramide fluid flows. They suggest that the overlapping late δD values for the early talc and magnesite ores record overprinting during the syn- to post- Laramide event. The counterpoint is that it is unreasonable to suggest that the overprinting event could be so highly selective that none of the early dolomite or Mississippi Valley-type ores were similarly affected.

One clear difference reported by Nesbitt and Muehlenbachs (1995a) is that their early population has primary two-phase fluid inclusions with low eutectic temperatures of −50° to −40°C and final melting temperatures of −25° to −15°C which indicate the presence of Ca ± Mg and Na ions in solution with a 20 to 25 wt percent NaCl equiv. In contrast the late vein population has only single-phase fluid inclusions with final melting temperatures of −25° to −20°C which indicate the presence of only Na ions with a 0 to 10 wt percent NaCl equiv. Also, they observe that the $^{87}Sr/^{86}Sr$ ratios of 0.7095 ± 0.0005 found in the Cambrian limestones in the southern Rocky Mountains, although more radiogenic than typical Cambrian values of 0.7090 ± 0.0002, are still distinctly lower than the 0.7097 ± 0.0017 values found for the saddle dolomites and Mississippi Valley-type ores in the Cambrian limestones or Devonian reefs of the western Canada sedimentary basin, thus favoring a late Paleozoic age. Further discussion of this data follows.

Pine Point Mississippi Valley-type deposit

Evidence from the Pine Point area has been central to the discussion of fluid flow in the western Canada sedimentary basin since Garven (1985) suggested that Laramide flow from emergence of the Rocky Mountains drove brines out of the western Canada sedimentary basin through the Presqu'ile reef aquifer and into the platform carbonates to form the Pine Point Mississippi Valley-type deposits over a 0.5 to 5 m.y. period in Paleocene time (Fig. 1). These deposits have been the most productive Mississippi Valley-type mines in Canada so that their genesis has been much discussed (Rhodes et al., 1984).

Cumming et al. (1990) tried to date galenas from Pine Point using lead isotope ratios but they proved very homogeneous so that, depending on the interpretation model, they obtained ages of ~60, ~300, ~375, or ≥386 Ma. Qing and Mountjoy (1992, 1994) argued from petrographic and stratigraphic evidence that the white hydrothermal dolomite must postdate deposition of the late Middle Devonian Slave Point

Formation, i.e., ≤375 Ma. From apatite fission track analyses of outcrop and well samples from a large area of the adjacent western Canada sedimentary basin and Precambrian Shield, Arne (1991) concluded that the host rocks at Pine Point were heated slowly by burial beginning in the Devonian, and that the 400-m-thick Paleozoic section reached a maximum paleotemperature of ~85° to 100°C upon maximum burial of ≤3 km during the Late Cretaceous. Recent apatite fission track thermochronometry on the east side of the western Canada sedimentary basin by Ravenhurst et al. (1994) supports these values and points to a Paleocene age. This event accounts for the ~85° to 100°C temperatures recorded in fluid inclusions in the ore (Roedder, 1968; Kyle, 1981). Thereafter rapid uplift led to the removal of ~2 km of overlying sedimentary rocks and to cooling by the mid-Tertiary. Arne (1991) noted that the preservation of pre-Devonian fission tracks in underlying rock units militates against a late Paleozoic age for the ores. Next, Qing and Mountjoy (1992) reported that the $^{87}Sr/^{86}Sr$ ratios decrease from 0.7106 to 0.7081, the homogenization temperatures decrease from 178° to 92°C, and the $\delta^{18}O$ values increase slightly from −16 to −7 per mil, going from the southwest end of the Presqu'ile reef to Pine Point at the northeast end. They attributed these trends to the eastward updip flow of hotter radiogenic basinal brines with the progressive admixture of cooler formational waters. They argued from known burial depths, fluid inclusion temperatures, maturation temperatures, etc., that this could have happened only in the Late Cretaceous to early Tertiary during the Laramide orogeny.

Next, Nakai et al. (1993) reported a 361 ± 13 Ma Rb-Sr sphalerite date for Pine Point, as discussed in the next section. In contrast, Symons et al. (1993) reported a Late Cretaceous or Paleocene age from paleomagnetism. Because the Pine Point dolomites and mineralization yielded very weak characteristic remanent magnetization intensities, they used less severe acceptance criteria than normal for the Kirschvink (1980) method. Nesbitt and Muehlenbachs (1995b) deemed the result to be unreliable on this basis, although the sole consequence is that the pole's cone of 95 percent confidence is a little larger. Clearly it provides a good fit for Late Cretaceous to early Tertiary time, with no evidence of any earlier age (Fig. 7b).

Nesbitt and Muehlenbachs (1994a) asserted that their geochemical data invalidated Garven's (1985) hydrogeological model. Garven and Sverjensky (1994) rebutted that Laramide compression on the west side of the western Canada sedimentary basin would drive brines with an early Paleozoic geochemistry first out of the basin and eastward into the platform at Pine Point, thereby explaining their data. In their reply Nesbitt and Muehlenbachs (1994b) suggested that vitrinite reflectance values (R_o) of 4.50 percent and fluid inclusion temperatures of 150° to 200°C for the Manetoe dolomite (Morrow et al., 1993) indicate that hot geothermal brines were present near surface in the Devonian to form coarse crystalline dolomite. These values from the west side of the Liard basin lead Morrow and Aulstead (1995) to support a late Paleozoic age for the Manetoe dolomite. However, the closest sample site of Morrow et al. (1993) to Pine Point is the Imperial Island River 1 well in the interior plains, about halfway across the western Canada sedimentary basin. Morrow et al. (1993) concluded that for it, pre-Cretaceous history had little effect on maturation profiles, and therefore, that it provided no evidence of a hot Devonian brine in the central or eastern part of the western Canada sedimentary basin.

In their comment on the Nesbitt and Muehlenbachs (1994a) paper, Qing and Mountjoy (1995) argued that the δD values for dolomites in Cambrian units of the southern Rocky Mountains and the western Canada sedimentary basin and in the Pine Point dolomite were not caused by an influx of late Paleozoic meteoric waters, but rather that they reflected a shift of the study area from a tropical paleolatitude during the Devonian to a subpolar paleolatitude by Cretaceous time. They suggested further that the high fluid inclusion salinities in the dolomites, and the shift in values for $\delta^{18}O$, $\delta^{13}C$, $^{87}Sr/^{86}Sr$, and fluid inclusion microthermometry could be best interpreted by the progressive mixing of meteoric water during the eastward flow of basinal brine during and after the Laramide orogeny. Nesbitt and Muehlenbachs (1995b) countered by postulating that high mountains to the west of the Rocky Mountains and the western Canada sedimentary basin could have led to depleted δD and $\delta^{18}O$ values. Viets (1995) found that the fluid inclusions in sphalerite at Pine Point have a distinctly bimodal geochemistry, raising the possibility of two fluid flow events at Pine Point.

Rb-Sr sphalerite ages at Pine Point

Nesbitt and Muehlenbachs (1993, 1995a) placed great weight on the Devonian Rb-Sr sphalerite age of 361 ± 13 Ma for Pine Point of Nakai et al. (1993). Since then J. Brannon (pers. commun., 1995) has obtained an Rb-Sr sphalerite age of 374 ± 21 Ma. However, the method itself is poorly understood with many inherent problems (Brannon et al., 1995). Garven and Sverjensky (1994) noted that the method is prone to contamination by colloidal clay particles entrapped in sphalerite because of the low concentration levels of the measured isotopes, so that even the slightest trace of clay will result in an older age that approaches or exceeds the host-rock age. Recently, Pettke and Diamond (1996) proposed a model for the method based on the differential partitioning of Rb and Sr between the hydrothermal fluid and sphalerite. Since most sphalerites grow inhomogeneously, they suggest that their fluid inclusions retain different Rb-Sr ratios to record a valid isochron. However, for example, Nakai et al. (1993) reported an impossible Rb-Sr age of ~600 Ma for sphalerite in Lower Ordovician rocks of the northern Arkansas Mississippi Valley-type district. For Pine Point, the 361 ± 13 and 374 ± 21 Ma Rb-Sr sphalerite ages substantially overlap the late Givetian age (375 ± 2 Ma) of the host Slave Point Formation (Nakai et al., 1993; J. Brannon, pers. commun., 1995). At Blendevale in Australia, the sphalerite ore in Late Devonian host rocks (370 ± 4 Ma) gives an Rb-Sr age that is only slightly younger at 357 ± 3 Ma, although it is consistent with indirect dating evidence from carbonate cement stratigraphy and the thermal history for the area (Christensen et al., 1995). For East Tennessee, Nakai et al. (1990, 1993) reported Rb-Sr sphalerite ages of 347 ± 20 and 377 ± 29 Ma which are statistically different. All of these values could reflect contamination by diagenetic or older clays. Fur-

ther, the Rb-Sr ages at Pine Point, Blendevale, and East Tennessee have not yet been supported by any other direct dating method, either radiometric or paleomagnetic. For the Newfoundland zinc Mississippi Valley-type deposits, the Rb-Sr sphalerite age of 375 ± 75 Ma (Nakai et al., 1993) agrees with the paleomagnetic age of 380 ± 7 Ma (Pan and Symons, 1993) but has an unacceptably large error. Only for the Polaris Mississippi Valley-type deposit in the Arctic Islands does the Rb-Sr sphalerite age of 366 ± 15 Ma agree with an earlier direct date, a paleomagnetic age of 367 ± 7 Ma (Symons and Sangster, 1992; Christensen et al., 1995). Clearly there are experimental problems yet to be resolved before Rb-Sr sphalerite ages can be accepted uncritically.

Paleomagnetism

In addition to the Pine Point study (Symons et al., 1993), four recent paleomagnetic studies provide important evidence for Laramide fluid flow on a large scale in the western Canada sedimentary basin. Symons et al. (1997b) report results from the host rocks near the Mount Brussilof magnesite deposit. Although the deposit itself failed to retain a coherent remanence, the adjacent dolostones of the Middle Cambrian Cathedral and Eldon Formations gave a tightly constrained Late Cretaceous to Paleocene pole based on data from 223 specimens from 11 sites (Fig. 7b). This remanence is shown to postdate Laramide folding by overwhelmingly negative paleomagnetic fold and tectonic breccia tests, as well as an antiparallel reversed polarity in a late-stage dolomite cement which must postdate the end of the Cretaceous normal superchron at ~84 Ma. Their results show that the Cambrian carbonates near Mount Brussilof on the immediate east side of the Cathedral escarpment were also recrystallized by hydrothermal fluid flow in the Laramide orogeny.

Enkin et al. (1997) reported paleomagnetic data from folded Upper Devonian to upper Mississippian limestones, dolostones, and silty dolostones at two locations along the Crowsnest Pass south of Mount Brussilof in the Main and Front Ranges and at one location just west of Calgary in the Foothills (Fig. 1). They found a pre- to syndeformational remanence at all three locations, marking the start of Laramide deformation. The paleomagnetic ages for the three areas are not yet tightly constrained because each is based on five or fewer sites and because the age limits are poorly known for the folding. Nevertheless, they find a Jurassic to Early Cretaceous age in the Main Range, a Cretaceous age (118–74 Ma) in the Front Range, and Late Cretaceous or younger age (≤83 Ma) in the Foothills. Their results show that the Paleozoic strata were profoundly affected by fluid flow in the early stages of the Laramide orogeny as deformation progressed northeastward into the margin of the western Canada sedimentary basin. Further, in an abstract, Osadetz et al. (1996b) reported that 105 paleomagnetic sites over a 400- by 100-km portion of the foreland belt yield normal and reverse polarity characteristic remanent magnetization directions of chemical origin in the Front Range and Foothills, respectively, and that they relate to the terminal stages of Laramide thrust faulting. Finally, Lewchuk et al. (1996) have reported paleomagnetic results from drill cores in the dolomitized Mississippian reef of the Shell Waterton gas field in the western Canada sedimentary basin. Except for a few specimens from an anhydrite-rich zone which preserve the Mississippian direction, all specimens give a Laramide characteristic remanent magnetization.

Thus, paleomagnetic dating at numerous places throughout the western Canada sedimentary basin requires extensive chemical alteration of the Paleozoic strata during Laramide orogenesis, supporting a pervasive Laramide fluid flow event.

Orogenesis

It would require kilometer(s)-scale uplift on the southwestern side of the western Canada sedimentary basin to drive fluids ~700 km across to Pine Point (Garven, 1985). Yet there is virtually no evidence for a late Paleozoic to pre-Laramide (pre-Late Jurassic) tectonism in the platform sequence of the foreland belt, such as marked angular unconformities or two generations of folding. Thus, both Yao and Demicco (1995) and Nesbitt and Muhlenbachs (1995b) suggest that orogenic uplift occurred farther west. The former relate dolomitization to uplift in the Kootenay arc from Late Silurian to Late Devonian as proposed by Okulitch (1985). The latter appeal to a Late Devonian to Carboniferous fluid flow event from the Antler orogeny, which may have uplifted the Purcell and Kootenay terranes in Middle to Late Devonian time. Both proposals are difficult to accept on stratigraphic evidence alone that requires the Mississippi Valley-type mineralization to postdate late Late Devonian time. Further, Nesbitt and Muehlenbachs (1995a) have shown that fluids flowing eastward from the Purcell anticlinorium and Omineca terrane and into the Foothills belt should result in significant changes in the temperatures and salinities of fluid inclusions, in the $\delta^{18}O$, δD, and $\delta^{13}C$ values but not in the $^{87}Sr/^{86}Sr$ values. These predictions are poor matches for the determined values summarized earlier for the early hydrothermal dolomite sheets relative to the late dolomite veins. In most cases, the spectra of values for the two populations overlap substantially, consistent with a fluid flow having an evolving chemistry rather than with two distinct episodes. Cogent explanations have been given by others for the geochemical data in the western Canada sedimentary basin in terms of an evolving Laramide to post-Laramide fluid flow (e.g., Garven and Sverjensky, 1994; Qing and Mountjoy 1995). Finally, recent fluid inclusion evidence shows that formational fluids evolve continuously during orogenic heating and burial up to median grades of metamorphism, but that the fluids retain distinct salinity characteristics inherited from their sedimentary precursors (Yardley et al., 1997). This evidence indicates that fluid inclusions in the hydrothermal dolomites should be expected to inherit Paleozoic characteristics from the carbonate platform strata if they were formed during the Laramide orogeny, and that this Paleozoic signature should wane in younger dolomites as the Laramide orogeny wanes as found by Nesbitt and Muehlenbachs (1995b).

Conclusions

The Mississippi Valley-type mineralization and carbonate host rocks of the Kicking Horse and Monarch mines retain a characteristic remanent magnetization which resides in both pyrrhotite and magnetite. Saturation isothermal remanent

magnetization analyses show that these remanence carriers have single domain to pseudosingle domain characteristics. Thermal step demagnetization analyses combined with published fluid inclusion and other petrologic data require that the characteristic remanent magnetization be chemical in origin. Paleomagnetic fold tests on a Laramide fold of an amplitude of a few meters and on a regional Laramide fold show that the characteristic remanent magnetization postdates most of this deformation and, therefore, that it is Late Cretaceous or younger in age. This observation is entirely consistent with the observations from mine mapping and from mapping of Mississippi Valley-type showings elsewhere in the southern Rocky Mountains. Also, the increase in characteristic remanent magnetization intensity in the mineralized zones relative to remote host rock indicates that the mineralization and magnetization events are coeval. Comparison of the characteristic remanent magnetization directions for the dolostones, the hydrothermal dolomites, and the mineralized zones in the Kicking Horse mine and near it on the northern side of the Kicking Horse Pass show that they are colinear, indicating that they acquired their characteristic remanent magnetizations coevally. When combined, they give a pole position that coincides with the mid-Early Cretaceous to Paleocene stillstand in the apparent polar wander path for North America. The few sites on the south side close to the Monarch mine give a pole position that is closest to this part of the apparent polar wander path also, and its minor deviation from the path is attributed to post-Laramide tectonic adjustments caused by sampling too close to a major fault. Finally, the overwhelming preponderance of normal polarity characteristic remanent magnetization directions indicates that the characteristic remanent magnetization was acquired mostly during the Cretaceous normal superchron which ended at ~84 Ma. These results show that the hydrothermal dolomite sheets and associated Mississippi Valley-type deposits were formed with a primary chemical characteristic remanent magnetization during Late Cretaceous Laramide uplift and deformation. It follows, therefore, that Laramide structures will provide one control for Mississippi Valley-type mineralization in the southern Rocky Mountains. The results also show that the Cambrian dolostone host rocks were either limestones which were dolomitized to acquire a primary remanence or dolostones which were recrystallized to acquire a reset secondary remanence during the Late Cretaceous.

Review of the evidence to date leads to the conclusion that the most significant fluid flow event in the western Canada sedimentary basin occurred in response to the Laramide orogeny. The stratigraphic, structural, paleomagnetic, fission track, hydrogeologic modeling, and most other cited evidence supports this statement. Also, the available fluid inclusion, geochemical, and isotopic evidence can be rationally explained in terms of an evolving fluid in the western Canada sedimentary basin from the peak of the Laramide orogeny to the present. Earlier fluid flow events may have occurred; however, if they were present then the evidence to date suggests that they were probably minor by comparison and have been erased by the overwhelming Laramide event.

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